

Dummit and Foote 7.1

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1 Problems

Prop 1. (1) $0a = (0 + 0)a = 0a + 0a$, so $0a = 0$ for all $a \in R$. Similarly, $a0 = a(0 + 0) = a0 + a0$, so $a0 = 0$ for all $a \in R$.

(2) $(-a)b + ab = (-a + a)b = 0b = 0$. Hence, $-(ab) = (-a)b$ for all $a, b \in R$. Similarly, $a(-b) + ab = a(-b + b) = a0 = 0$, so $-(ab) = a(-b)$ for all $a, b \in R$.

(3) $(-a)(-b) = -(a(-b)) = -(-ab) = ab$ for all $a, b \in R$.

(4) $1 = 11' = 1'$, so if the identity exists, it is unique. From proposition 2, $(-1)a = -(1a) = -a$ for all $a \in R$.

The set of units $R^\times$ forms a group. $11 = 11 = 1$, so $1 \in R^\times$. If $a \in R^\times$ so that there exists $ab = ba = 1$, $b \in R^\times$ as well. This means inverses exist in $R^\times$. Associativity is inherited from the associativity of $\cdot$ of R . $R^\times$ is closed under $\cdot$ as $ab(b^{-1}a^{-1}) = b^{-1}a^{-1}ab = 1$.

A zero divisor cannot be a unit. Suppose the contrary that z is a zero divisor and unit of R . If $zw = 0$ for some $w \in R - \{0\}$, pick z^{-1} such that $z^{-1}z = 1$. Then $w = z^{-1}zw = z^{-1}0 = 0$, or $w = 0$, contradiction. If $wz = 0$, $w = wzz^{-1} = 0z^{-1} = 0$.

Every element of $\mathbb{Z}/n\mathbb{Z}$ is either a zero divisor or unit. This property also holds for the ring of functions from $[0, 1]$ to $\mathbb{Z}$.

Prop 2. Let $a, b, c \in R$ such that $ab = ac$ and a is not a zero divisor. Then $a = 0$ or $b = c$. This holds always if R is an integral domain.

Write $a(b - c) = 0$. Either $a = 0$ or $a \neq 0$, for which $b - c = 0$.

Corro 3. Any finite integral domain is a field.

It suffices to show the existence of inverses. Consider the set $R' = R - \{0\}$ and $aR' = \{ar : r \in R'\}$ for some nonzero a . Suppose $ar = as$. Then $a = 0$

or $r = s$. By assumption, $r = s$, for which the elements of aR' are distinct. R and R' are the same size, so there exists a^{-1} for which $aa^{-1} = 1$. Hence, R is a field.

Let D be a square-free integer such that $D \equiv 1 \pmod{4}$. Show that $R = \mathbb{Q}[\frac{1+\sqrt{D}}{2}]$ is a ring.

Let $A = a + b\frac{1+\sqrt{D}}{2}, X = x + y\frac{1+\sqrt{D}}{2} \in R$. Then

$$\begin{aligned} AX &= ax + (ay + bx)\frac{1+\sqrt{D}}{2} + by\frac{D+2\sqrt{D}+1}{4} \\ &= \left(ax + by\frac{D-1}{4}\right) + (ay + bx + by)\frac{1+\sqrt{D}}{2}. \end{aligned}$$

Let $\alpha = a + b\frac{1+\sqrt{D}}{2}$. Compute $N(\alpha)$.

$$\begin{aligned} N(\alpha) &= N\left(\left(a + \frac{b}{2}\right) + \frac{b}{2}\sqrt{D}\right) \\ &= \left(a + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 D \\ &= a^2 + ab + \frac{1-D}{4}b^2, \end{aligned}$$

so $N(\alpha) \in \mathbb{Z}$ for all α .

$N(\alpha) = \pm 1$ iff α is a unit of R .

Suppose $N(\alpha) = 1$. Then $\bar{\alpha}\alpha = \alpha\bar{\alpha} = 1$, so $\alpha \in R^\times$. The case for $N(\alpha) = -1$ is similar. Now suppose $\alpha\beta = \beta\alpha = 1$. Then $N(\alpha)N(\beta) = N(1) = 1$. Since $N(\alpha), N(\beta) \in \mathbb{Z}$, we have $N(\alpha) = \pm 1$, AWD.

The units of $\mathbb{Q}\sqrt{D}$. First case, $D < 0$ and $D \equiv 2, 3 \pmod{4}$. We have $x^2 - Dy^2 = -1$. If $D \neq -1$ and $y \neq 0$, then $x^2 - Dy^2 > 1$. The only units are ± 1 . When $D = -1$, the units (of the Gaussian integers) are $\pm 1, \pm i$. Now suppose $D \equiv 1 \pmod{4}$. We want $(2a+b)^2 - Db^2 = 4$. If $D < -4$ and $b \neq 0$, then $-Db^2 > 4$ and the units are ± 1 . We are left with $D = -3$. If $b = 0$, $a = \pm 1$. If $b = 1$, $2a + 1 = \pm 1$, so $a = 0, -1$. If $b = -1$, $a = 0, 1$. This gives the six roots of unity as units.

The case where $D > 0$ is separate and depends on Pell's equations.

P 7. Let $z, w \in Z(R)$ and $r \in R$. Then

$$(z + w)r = zr + wr = rz + rw = r(z + w),$$

so $z + w \in Z(R)$ as well. $zwr = zrw = rzw$, so $zw \in R$ as well. $1r = r1 = r$, so if $1 \in R$, $1 \in Z(R)$ as well. $(-z)r = -(zr) = -(rz) = r(-z)$, so $-z \in Z(R)$ as well. $0 \in Z(R)$, so $Z(R) \neq \emptyset$.

Suppose R is a division ring. For any $a \neq 0 \in Z(R)$, there exists $b \in R$ such that $ab = ba = 1$. Note $br = brab = abrb = rb$, so $b \in Z(R)$ as well. $Z(R)$ is closed under inverses, AWD (since $Z(R)$ is a commutative division ring).

P 14. Let x be a nilpotent element of a commutative ring R .

(a) We proceed by induction on m . If $m = 2$, $x^2 = 0$. If $x \neq 0$, then x is a zero divisor. Else, $x = 0$ by assumption. Assume the result for $m = k$. Consider the equation $xx^k = x^{k+1} = 0$. If $x \neq 0$ and $x^k \neq 0$, then x is a zero divisor. Else $x \neq 0$ and $x^k = 0$, for which we apply the IH. Else, $x = 0$ by assumption, AWD.

(b) $(rx)^m = r^m x^m = 0$.

(c) Take m to be even. Then $(1+x)(1-x+x^2-\dots-x^{m-1}) = 1-x^m = 1$.

(d) Let x, u be a nilpotent and unit element.

$$u+x = u(1+u^{-1}x).$$

$u^{-1}x$ is nilpotent, $1+u^{-1}x$ is a unit, $u(1+u^{-1}x)$ is a unit.

P 15. First, $a = a^2 = (-a)^2 = -a$. Second,

$$a+b+ab+ba = a^2+b^2+ab+ba = (a+b)^2 = a+b,$$

so $ab = ba$, AWD.

P 21. (a) The operation $\oplus$ is commutative and associative (by considering associativity of $\oplus$ on booleans). $\emptyset \oplus A = A \oplus \emptyset = A$. Moreover, $A \oplus A = \emptyset$, so $(\mathcal{P}, \oplus)$ defines an Abelian group. $A \cap B \in \mathcal{P}$ and we wish to show $(A \oplus B) \cap C = (A \cap C) \oplus (B \cap C)$. This may be expanded by De Morgan's Laws and the definition of $\oplus$.

(b) $\cap$ is commutative. $X \cap A = A \cap X = A$, and $A \oplus A = \emptyset$ as we see above. $\mathcal{P}$ is a boolean ring.

P 22. Take X to be infinite, where X is the set in P 21.

P 29. If $f, g \in R$, $x, y \in A$, then

$$(f+g)(x+y) = (f(x)+g(x)) + (f(y)+g(y)) = f(x+y) + g(x+y),$$

so $f+g \in R$.

$$f(g(x+y)) = f(g(x)+g(y)) = f(g(x)) + f(g(y)),$$

so $f \circ g \in R$. The identity of R is $\text{id}(A)$, since $f \circ \text{id} = \text{id} \circ f = f$. Function composition is associative, so R is indeed a ring. If $u \in R$ is a unit, there exists $v \in R$ for which $u \circ v = v \circ u = \text{id}$. This means u is a bijective homomorphism, i.e. an automorphism of A .